UNIT -V

Knowledge Management Metrics, Organizational Culture-Types and analysis, Organizational maturity model, Artificial Intelligence and Expert Systems: Concepts and Definitions of Artificial Intelligence, Artificial Intelligence Versus Natural Intelligence, Machine Learning-Data Distribution, Machine Learning Process, Tools, TensorFlow

Knowledge Management Metrics :

Knowledge Management (KM:

Knowledge Management (KM) is the process of collecting, sharing, and using the knowledge and expertise within an organization to improve its efficiency and decision-making.

Key Points:

1. **Knowledge Sharing:** Sharing important information and ideas among employees, teams, or departments.
2. **Knowledge Creation:** Developing new knowledge through research, collaboration, or learning.
3. **Knowledge Storage:** Storing valuable knowledge in easy-to-access places, like databases or files, so that people can find and use it when needed.
4. **Knowledge Application:** Using the shared knowledge to solve problems, make decisions, and improve work.
5. **Knowledge Retention:** Keeping important knowledge in the organization even when employees leave or retire.

Simple Example:

Imagine a company where employees frequently solve problems by asking each other for help. Over time, they start collecting solutions to common problems in a shared online folder. New employees can access this folder to find answers to their questions without needing to ask others. This helps save time, avoid mistakes, and keeps the company running smoothly. This is an example of **Knowledge Management** in action.

Knowledge Management Metrics :

KM metrics are tools used to measure how well Knowledge Management (KM) activities are working. These metrics help an organization understand the effectiveness of its KM practices and where improvements can be made.

1. **Knowledge Utilization:**
   - **What it measures:** How often knowledge is accessed, shared, or applied by employees.
   - **Simple Example:** In a company, there's a shared document with troubleshooting tips. If employees frequently open and use this document to solve problems, it shows good knowledge utilization.
2. **Employee Collaboration:**
   - **What it measures:** The frequency and quality of collaboration between employees or teams on knowledge-sharing platforms.
   - **Simple Example:** Employees regularly post questions and answers on a company's internal chat system. This means they are actively collaborating and sharing knowledge.
3. **Knowledge Retention Rate:**
   - **What it measures:** The percentage of important knowledge that is retained in the organization over time, even when employees leave.
   - **Simple Example:** When a senior employee retires, their knowledge is transferred to a new employee through training or documentation, ensuring that the organization doesn't lose important know-how.
4. **Innovation and Improvement:**
   - **What it measures:** How knowledge sharing and application contribute to new ideas or improved business processes.

    - **Simple Example:** A company shares market research reports across teams, leading to the development of a new product based on insights gained. This shows how knowledge sharing sparks innovation.
5. **Training and Development:**
    - **What it measures:** How many training sessions on KM tools and strategies are conducted, and how well employees use them.
    - **Simple Example:** The company holds training on a new knowledge-sharing software, and employees start using it to share their expertise and find information faster. This shows how the training helped improve KM practices.
6. **Customer Satisfaction:**
    - **What it measures:** How well internal knowledge improves customer service and satisfaction.
    - **Simple Example:** A customer service team uses a knowledge base to quickly find solutions to common customer issues. This results in faster service and happier customers.

These metrics help the organization measure the success of its Knowledge Management efforts and find areas where it can improve.

<u>Role of People in Knowledge Management Metrics :</u>

1. Knowledge Utilization

- **What it measures**: How often knowledge is accessed, shared, or applied by employees.
- **Role of People**: Employees actively engage with shared knowledge resources (documents, knowledge bases, etc.). Their willingness to access and apply knowledge directly impacts this metric.
    - **Example**: In a company, the troubleshooting document is used regularly by employees to solve problems. The people who are frequent users of the document demonstrate knowledge utilization in action. If they find the document useful and often turn to it when encountering problems, it highlights how effective the knowledge is in addressing their needs.

2. Employee Collaboration

- **What it measures**: The frequency and quality of collaboration between employees or teams on knowledge-sharing platforms.
- **Role of People**: Employees need to actively participate in knowledge-sharing platforms (e.g., internal chat systems, forums, or collaborative software). Their interactions (posting questions, answering queries, sharing insights) directly contribute to this metric.
    - **Example**: If employees regularly use an internal chat system to exchange information, the people involved (those asking and answering questions, sharing expertise) are contributing to improved collaboration. The quality of their contributions—how well the information is shared, understood, and applied—affects the overall effectiveness of collaboration.

3. Knowledge Retention Rate

- **What it measures**: The percentage of important knowledge that is retained in the organization over time, even when employees leave.
- **Role of People**: People (especially senior employees) are critical in knowledge transfer. Through mentoring, documentation, or training, they ensure that valuable knowledge doesn't leave when they exit the company.
    - **Example**: When a senior employee retires, the responsibility falls on both the retiring employee and the receiving employee to ensure the transfer of knowledge through training sessions, documentation, or shadowing. The success of this transfer, led by the involved individuals, impacts the retention rate of critical knowledge.

4. Innovation and Improvement

- **What it measures**: How knowledge sharing and application contribute to new ideas or improved business processes.
- **Role of People**: Employees who share insights and contribute their knowledge foster an environment ripe for innovation. By sharing experiences, data, and research, they create the foundation for new ideas and business process improvements.

    - **Example**: When market research insights are shared across teams, individuals can apply those insights to their respective areas. The collaboration and knowledge shared by those employees lead to the creation of a new product or an improvement in an existing process. The people who contribute this knowledge play a direct role in driving innovation and improvement.

5. Training and Development

- **What it measures**: How many training sessions on KM tools and strategies are conducted, and how well employees use them.
- **Role of People**: Employees are the recipients and participants in training. Their engagement and active use of newly learned tools directly affect how well KM practices are implemented and refined within the organization.
    - **Example**: After attending training on a new knowledge-sharing software, employees begin to actively use it to share documents, communicate insights, and ask questions. Their adoption and successful application of the tool reflect how well the training has prepared them to improve KM practices.

6. Customer Satisfaction

- **What it measures**: How well internal knowledge improves customer service and satisfaction.
- **Role of People**: Employees in customer-facing roles (e.g., customer service teams) are directly responsible for using internal knowledge to address customer needs efficiently and effectively. Their ability to leverage internal knowledge bases impacts the quality of service provided.
    - **Example**: When customer service representatives use a well-maintained knowledge base to quickly resolve customer issues, they can deliver faster and more accurate responses. The employees who are skilled at utilizing this resource ensure that customer satisfaction is improved by providing timely, accurate information.

Organizational Culture: Types and Analysis

Organizational culture refers to the values, beliefs, behaviors, and attitudes that define how things are done within a company. It's like the personality of the organization, shaping how employees interact, make decisions, and solve problems.

Types of Organizational Culture

1. **Clan Culture** (Family-like, Collaborative)
    - **Description**: Focuses on teamwork, loyalty, and collaboration. People work together like a family, with a strong emphasis on trust and support.
    - **Characteristics**: Friendly, supportive, open communication.
    - **Example**: A small startup where everyone shares ideas freely, helps each other out, and celebrates group achievements. Employees have a strong sense of belonging.
    - **Pros**: High employee morale, good teamwork, supportive environment.
    - **Cons**: Can be too informal, may lack clear structure.
2. **Adhocracy Culture** (Innovative, Creative)
    - **Description**: Focuses on innovation, creativity, and risk-taking. The company encourages employees to experiment and come up with new ideas.
    - **Characteristics**: Flexible, dynamic, forward-thinking.
    - **Example**: A tech company like Google, where employees are encouraged to think outside the box, take risks, and work on creative projects without many rules.
    - **Pros**: High innovation, quick adaptation to change.
    - **Cons**: Can be chaotic, lack of stability.
3. **Market Culture** (Competitive, Results-oriented)
    - **Description**: Focuses on achieving goals, winning, and being competitive. Employees are expected to perform at their best, and success is measured by results.
    - **Characteristics**: Goal-oriented, competitive, performance-driven.
    - **Example**: A sales-driven company where employees are rewarded based on performance, and the focus is on meeting targets and winning in the marketplace.
    - **Pros**: Clear goals, high performance.
    - **Cons**: Can create stress, employees may feel pressured.

4. **Hierarchy Culture** (Structured, Controlled)
    - **Description**: Focuses on stability, rules, and clear organizational structure. Employees are expected to follow procedures, and there's a clear hierarchy of authority.
    - **Characteristics**: Stable, organized, controlled.
    - **Example**: A government agency or a large corporation where there are lots of rules, formal communication channels, and employees follow established processes.
    - **Pros**: Clear roles, efficient processes, stability.
    - **Cons**: Can be rigid, slow to adapt to change.

Analyzing Organizational Culture

- **How to Analyze Culture**:
    - **Look at the work environment**: Is it relaxed or strict? Are employees working in teams or individually?
    - **Communication style**: Are ideas shared openly, or is communication more formal and top-down?
    - **Decision-making**: Who makes decisions? Is it a group effort, or do leaders make all the calls?
    - **Values and Beliefs**: What does the company emphasize—innovation, results, employee well-being, or adherence to rules?
    - **Behavioral Patterns**: How do employees act in their daily work? Are they motivated by competition, collaboration, or stability?

Example of Organizational Culture in Action:

- **Company A (Clan Culture)**: Employees regularly meet for team-building activities, and there's a strong sense of community. They focus on personal growth and internal development. The company is often seen as "family-like" where relationships matter just as much as work tasks.
- **Company B (Adhocracy Culture)**: Employees are given freedom to explore new ideas and are not afraid to take risks. They may fail, but the company encourages them to learn from mistakes and keep trying. There's a constant push for creative thinking and innovation.
- **Company C (Market Culture)**: This company is highly competitive, with clear performance metrics. Employees are motivated by bonuses based on performance and are always striving to meet targets and surpass competitors.
- **Company D (Hierarchy Culture)**: This company follows strict procedures. Employees have well-defined roles, and decisions are made by

top management. The focus is on following processes and maintaining stability.

Organizational maturity model:

An **Organizational Maturity Model** is a way to assess how well an organization performs certain processes and practices. It helps organizations understand where they are in terms of growth and improvement. Think of it like a "growth chart" for a business that shows how developed and efficient its systems and operations are.

It is typically broken down into **stages**, which help the organization see where they stand and what needs to be done to improve.

Common Stages in Organizational Maturity

1. **Initial (Ad-hoc)**:
   - **Description**: At this stage, the organization has no standardized processes. Everything is done on the fly, and people often work in silos without much coordination.
   - **Characteristics**:
     - Processes are unpredictable.
     - No clear structure or consistency.
     - Problems are solved on an as-needed basis.
   - **Example**: A small startup where everyone works independently, and decisions are made as problems come up. There are no formal procedures or plans.
2. **Managed (Repeatable)**:
   - **Description**: The organization begins to establish basic processes and systems, though they may not be fully optimized. There is more consistency, but processes might still vary.
   - **Characteristics**:
     - Some processes are documented.
     - Projects are more predictable, but there's still room for improvement.

        - People start using tools and techniques to manage tasks better.
    - **Example**: A company that starts using project management software to track tasks and deadlines but still faces inconsistencies in how projects are managed.
3. **Defined (Standardized)**:
    - **Description**: The organization has standardized processes that everyone follows. There's a clear structure and consistency across the organization.
    - **Characteristics**:
        - Processes are documented and followed by everyone.
        - There is a focus on efficiency and effectiveness.
        - Tools, techniques, and roles are clearly defined.
    - **Example**: A company with established procedures for customer service, HR processes, and project delivery. Everyone knows their role and follows a set process.
4. **Quantitatively Managed (Measured)**:
    - **Description**: At this stage, the organization uses data and metrics to measure and improve its processes. It is more analytical and uses performance data to drive decisions.
    - **Characteristics**:
        - Data and metrics are regularly tracked.
        - Performance is analyzed, and improvements are made based on numbers.
        - There's a strong focus on continuous improvement.
    - **Example**: A company that tracks customer satisfaction scores, employee performance, and financial data to improve decision-making and processes.
5. **Optimizing (Continuous Improvement)**:
    - **Description**: The organization has achieved a high level of maturity. Processes are constantly refined and improved, and there's a culture of innovation and learning.
    - **Characteristics**:
        - Focus on innovation and continuous improvement.
        - Processes are consistently reviewed and updated based on feedback.
        - The organization is flexible and adapts quickly to changes.
    - **Example**: A company that actively looks for ways to improve operations, keeps an eye on industry trends, and encourages employees to innovate and improve processes regularly.

Example of Organizational Maturity in Action:

Let's say a company is looking at its **customer service** process:

- **Initial (Ad-hoc)**: Customer service is handled by whoever is free. There's no clear process, and issues are handled as they come.
- **Managed (Repeatable)**: The company creates a basic customer service script and assigns specific people to handle customer inquiries, but there's still inconsistency.
- **Defined (Standardized)**: The company has a detailed customer service manual, and all employees follow the same steps to handle customer issues. It's efficient but could still be improved.
- **Quantitatively Managed (Measured)**: The company tracks metrics such as response times and customer satisfaction. They use this data to optimize the process and improve service.
- **Optimizing (Continuous Improvement)**: The company constantly reviews customer feedback, adjusts their process, and tries new technologies (like chatbots) to make customer service even better.

Key Benefits of Using an Organizational Maturity Model:

- **Understanding Strengths and Weaknesses**: The model helps identify where the organization excels and where it needs improvement.
- **Targeted Growth**: It helps focus on specific areas for development, rather than trying to improve everything at once.
- **Clear Progression**: The stages show a clear path for growth, guiding the organization from basic processes to optimized performance.

Artificial Intelligence and Expert Systems: Concepts and Definitions of Artificial Intelligence, Artificial Intelligence versus Natural Intelligence:

Artificial Intelligence (AI) :

Artificial Intelligence (AI) is the field of computer science that focuses on creating machines or software that can perform tasks that would typically require human intelligence. It enables computers to **think, learn, and make decisions** similar to humans, but with the ability to process large amounts of data quickly.

Key Points About AI:

1. **Machines That Learn**: AI systems can learn from data. They improve their performance over time as they are exposed to more information.
   - **Example**: A recommendation system on Netflix that learns your viewing preferences and suggests movies you might like based on what you've watched before.
2. **Mimicking Human Thinking**: AI aims to replicate human-like thinking, decision-making, and problem-solving abilities in machines.
   - **Example**: Virtual assistants like Siri or Alexa can understand your questions and provide answers, just like a human would.
3. **Automation**: AI can automate repetitive tasks, saving time and effort for humans. It can handle jobs that would take humans much longer to complete.
   - **Example**: In a factory, robots powered by AI can assemble products faster and more efficiently than humans.
4. **Decision-Making**: AI can analyze data and make decisions, sometimes even better than humans in certain situations, by recognizing patterns in data.
   - **Example**: AI in healthcare can analyze medical images and help doctors detect diseases like cancer early.
5. **Natural Language Processing (NLP)**: AI can understand and generate human language, enabling communication with computers in everyday language.
   - **Example**: Chatbots on websites that answer customer questions instantly, or Google Translate, which translates languages in real-time.

Examples of AI in Everyday Life:

1. **Smartphones**: AI powers features like facial recognition, voice assistants (Siri, Google Assistant), and predictive text on your phone.
2. **Online Shopping**: AI systems recommend products to you based on your browsing history and what others like you have bought.
   - **Example**: Amazon recommends items based on your past purchases and browsing behavior.
3. **Self-Driving Cars**: AI is used in autonomous vehicles to recognize objects, make decisions, and navigate the road without human input.
   - **Example**: Tesla cars use AI to drive themselves with minimal human interaction.
4. **Social Media**: AI determines which posts you see on platforms like Facebook and Instagram based on your interests and interactions.

- **Example**: Facebook's news feed shows you posts that the AI thinks you'll find interesting based on your past activity.

Artificial Intelligence Versus Natural Intelligence (Human Intelligence):

Aspect	Artificial Intelligence (AI)	Natural Intelligence (NI)
Origin	Created by humans through programming and algorithms.	Comes from living beings, particularly humans and animals.
Learning Method	Learns from data using machine learning and algorithms.	Learns through experiences, interactions, and sensory inputs.
Flexibility	Limited to specific tasks and struggles with generalization.	Highly flexible and adaptable to various tasks and environments.
Problem Solving	Solves problems based on logic, patterns, and predefined rules.	Uses intuition, emotions, and creative thinking to solve problems.
Creativity	Follows patterns and algorithms; lacks true creativity.	Can create entirely new ideas and solutions based on creativity.
Emotions	Does not have emotions or consciousness.	Humans and animals have emotions and feelings.
Speed	Can process data and make decisions extremely fast.	Humans process data more slowly but make thoughtful decisions.
Memory	Has perfect recall, stores large amounts of data, but needs to be programmed.	Human memory is imperfect and influenced by emotions or distractions.
Energy Consumption	Requires substantial computing power and energy.	More energy-efficient, uses natural brain processes.
Ethical Considerations	Can raise ethical concerns, lacks moral understanding.	Humans have ethics and morals to guide decisions.

Expert System :

5.11 Basic Concepts of Expert Systems

Q. 5.11.1	Explain basic concepts of expert systems. (Ref. Sec. 5.11)	**(5 Marks)**
Q. 5.11.2	What are expert systems? (Ref. Sec. 5.11)	**(5 Marks)**

Expert Systems (ES) are one of the prominent research domains of AI. It is introduced by the researchers at Stanford University, Computer Science Department.

The expert systems are the computer applications developed to solve complex problems in a particular domain, at the level of extra-ordinary human intelligence and expertise.

☛ Characteristics of Expert Systems

- High performance.
- Understandable.
- Reliable.
- Highly responsive.

☛ Capabilities of Expert Systems

The expert systems are capable of :

- Advising.
- Instructing and assisting human in decision making.
- Demonstrating.
- Deriving a solution.
- Diagnosing.
- Explaining.

- Interpreting input.
- Predicting results.
- Justifying the conclusion.
- Suggesting alternative options to a problem.

☞ In Capabilities of Expert Systems

They are incapable of :

- Substituting human decision makers.
- Possessing human capabilities.
- Producing accurate output for inadequate knowledge base.
- Refining their own knowledge.

Syllabus Topic : Structure of Expert Systems

5.12 Components of Expert Systems

Q. 5.12.1 Explain components of expert system. (Ref. Sec. 5.12) **(5 Marks)**

Q. 5.12.2 Explain structure of expert systems. (Ref. Sec. 5.12) **(5 Marks)**

The components of ES include :

- Knowledge Base.
- Inference Engine.
- User Interface.

Let us see them one by one briefly :

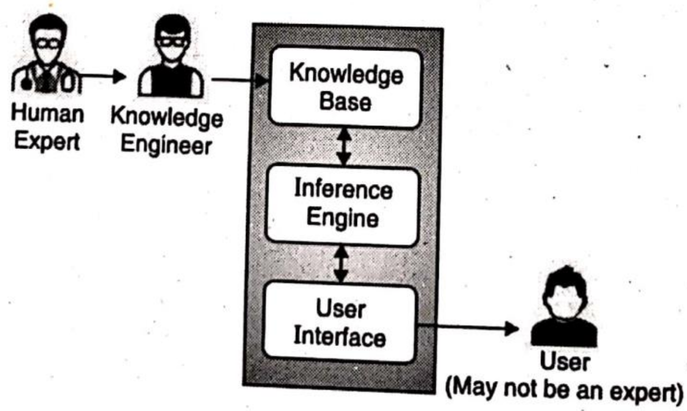

Fig. 5.12.1

Syllabus Topic : Knowledge Engineering

5.12.1 Knowledge Base

Q. 5.12.3 What is Knowledge? (Ref. Sec. 5.12.1) (5 Marks)

- It contains domain-specific and high-quality knowledge.
- Knowledge is required to exhibit intelligence. The success of any ES majorly depends upon the collection of highly accurate and precise knowledge.

The data is collection of facts. The information is organized as data and facts about the task domain. **Data, information,** and **past experience** combined together are termed as knowledge.

5.12.1.1 Components of Knowledge Base

Q.5.12.4 Explain forward chaining and backward chaining. (Ref. Sec. 5.12.1.1) (5 Marks)

The knowledge base of an ES is a store of both, factual and heuristic knowledge.

- **Factual Knowledge** : It is the information widely accepted by the Knowledge Engineers and scholars in the task domain.
- **Heuristic Knowledge** : It is about practice, accurate judgement, one's ability of evaluation, and guessing.

☞ **Knowledge representation**

It is the method used to organize and formalize the knowledge in the knowledge base. It is in the form of IF-THEN-ELSE rules.

☞ **Knowledge acquisition**

- The success of any expert system majorly depends on the quality, completeness, and accuracy of the information stored in the knowledge base.
- The knowledge base is formed by readings from various experts, scholars, and the **Knowledge Engineers**. The knowledge engineer is a person with the qualities of empathy, quick learning, and case analyzing skills.
- He acquires information from subject expert by recording, interviewing, and observing him at work, etc. He then categorizes and organizes the information in a meaningful way, in the form of IF-THEN-ELSE rules, to be used by interference machine. The knowledge engineer also monitors the development of the ES.

5.12.2 Inference engine

- Use of efficient procedures and rules by the Inference Engine is essential in deducting a correct, flawless solution.
- In case of knowledge-based ES, the Inference Engine acquires and manipulates the knowledge from the knowledge base to arrive at a particular solution.
- In case of rule based ES, it :
  - Applies rules repeatedly to the facts, which are obtained from earlier rule application.
  - Adds new knowledge into the knowledge base if required.
  - Resolves rules conflict when multiple rules are applicable to a particular case.
- To recommend a solution, the Inference Engine uses the following strategies :

1. Forward Chaining
2. Backward Chaining

→ 1. Forward Chaining

- It is a strategy of an expert system to answer the question, **"What can happen next?"**
- Here, the Inference Engine follows the chain of conditions and derivations and finally deduces the outcome. It considers all the facts and rules, and sorts them before concluding to a solution.
- This strategy is followed for working on conclusion, result, or effect. For example, prediction of share market status as an effect of changes in interest rates.

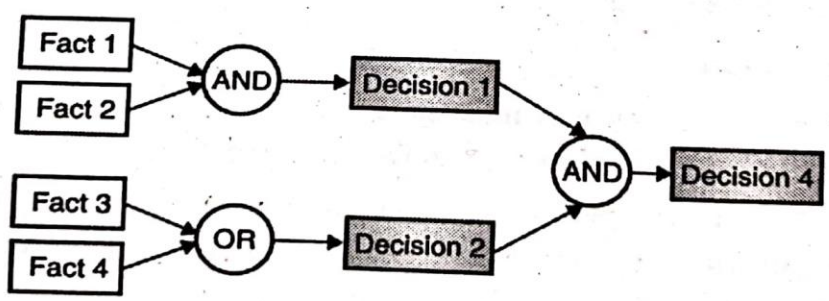

Fig. 5.12.2

→ 2. Backward Chaining

- With this strategy, an expert system finds out the answer to the question, **"Why this happened?"**

- On the basis of what has already happened, the Inference Engine tries to find out which conditions could have happened in the past for this result. This strategy is followed for finding out cause or reason. For example, diagnosis of blood cancer in humans.

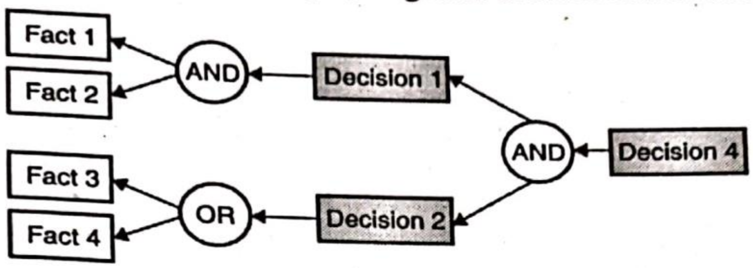

Fig. 5.12.3

5.12.3 User Interface

- User interface provides interaction between user of the ES and the ES itself. It is generally Natural Language Processing so as to be used by the user who is well-versed in the task domain.
- The user of the ES need not be necessarily an expert in Artificial Intelligence.
- It explains how the ES has arrived at a particular recommendation. The explanation may appear in the following forms :
  - Natural language displayed on screen.
  - Verbal narrations in natural language.
  - Listing of rule numbers displayed on the screen.
- The user interface makes it easy to trace the credibility of the deductions.

Requirements of Efficient ES user interface

- It should help users to accomplish their goals in shortest possible way.
- It should be designed to work for user's existing or desired work practices.
- Its technology should be adaptable to user's requirements; not the other way round.
- It should make efficient use of user input.

Expert systems limitations

No technology can offer easy and complete solution. Large systems are costly, require significant development time, and computer resources. ESs have their limitations which include :

- Limitations of the technology.
- Difficult knowledge acquisition.
- ES are difficult to maintain.
- High development costs.

Syllabus Topic : Applications of Expert Systems

5.13 Applications of Expert System

Q. 5.13.1 Explain applications of expert system in detail. (Ref. Sec. 5.13) **(5 Marks)**

The Table 5.13.1 shows where ES can be applied.

Table 5.13.1

Application	Description
Design Domain	Camera lens design, automobile design.
Medical Domain	Diagnosis Systems to deduce cause of disease from observed data, conduction medical operations on humans.
Monitoring Systems	Comparing data continuously with observed system or with prescribed behavior such as leakage monitoring in long petroleum pipeline.
Process Control Systems	Controlling a physical process based on monitoring.
Knowledge Domain	Finding out faults in vehicles, computers.
Finance/Commerce	Detection of possible fraud, suspicious transactions, stock market trading, Airline scheduling, cargo scheduling.

5.13.1 Expert System Technology

Q. 5.13.2 Write application of expert system. (Ref. Sec. 5.13.1) (5 Marks)

There are several levels of ES technologies available. Expert systems technologies include :

1. Expert System Development Environment

- The ES development environment includes hardware and tools.

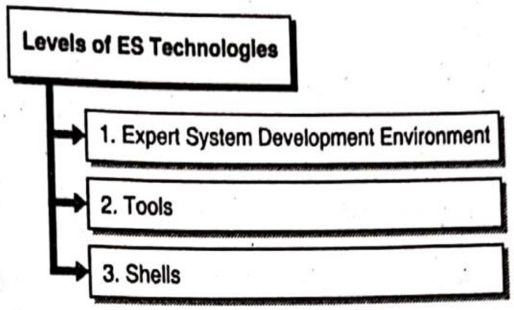

Fig. 5.13.1 : Levels of ES Technologies

They are :

- o Workstations, minicomputers, mainframes.
- o High level Symbolic Programming Languages such as **LIS**t **P**rogramming (LISP) and **PRO**grammation en **LOG**ique (PROLOG).
- o Large databases.

2. Tools

- They reduce the effort and cost involved in developing an expert system to large extent.
  - o Powerful editors and debugging tools with multi-windows.
  - o They provide rapid prototyping.
  - o Have Inbuilt definitions of model, knowledge representation, and inference design.

3. Shells

- A shell is nothing but an expert system without knowledge base. A shell provides the developers with knowledge acquisition, inference engine, user interface, and explanation facility. For example, few shells are given below :
  - o Java Expert System Shell (JESS) that provides fully developed Java API for creating an expert system.
  - o *Vidwan*, a shell developed at the National Centre for Software Technology, Mumbai in 1993. It enables knowledge encoding in the form of IF-THEN rules.

Syllabus Topic : Development of Expert Systems

5.14 Development of Expert Systems: General Steps

Q. 5.14.1 Enlist and explain steps of development of expert system. (Ref. Sec. 5.14) (5 Marks)

The process of ES development is iterative. Steps in developing the ES include :

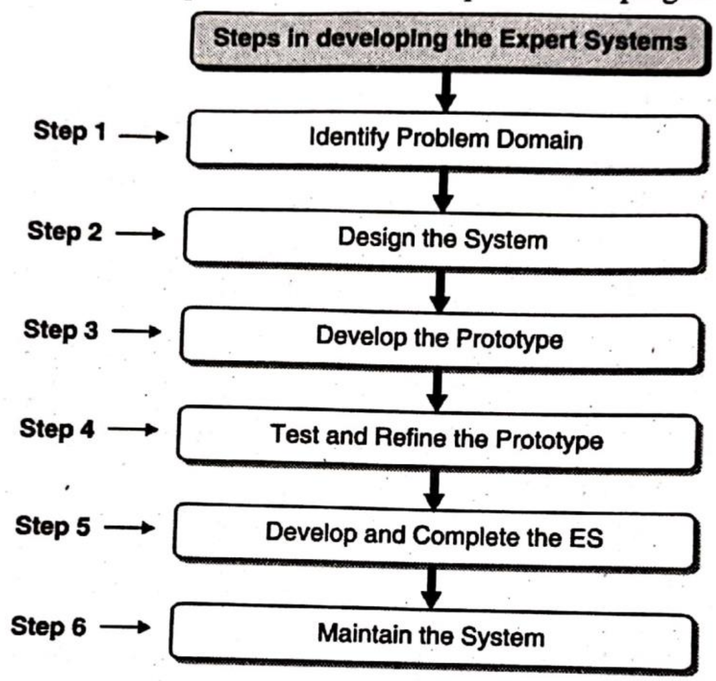

Fig. 5.14.1 : Steps in developing the Expert Systems

➜ 1. Identify Problem Domain

- The problem must be suitable for an expert system to solve it.
- Find the experts in task domain for the ES project.
- Establish cost-effectiveness of the system.

➜ 2. Design the System

- Identify the ES Technology.
- Know and establish the degree of integration with the other systems and databases.
- Realize how the concepts can represent the domain knowledge best.

3. Develop the Prototype

From Knowledge Base: The knowledge engineer works to :

- Acquire domain knowledge from the expert.
- Represent it in the form of If-THEN-ELSE rules.

4. Test and Refine the Prototype

- The knowledge engineer uses sample cases to test the prototype for any deficiencies in performance.
- End users test the prototypes of the ES.

5. Develop and Complete the ES

- Test and ensure the interaction of the ES with all elements of its environment, including end users, databases, and other information systems.
- Document the ES project well.
- Train the user to use ES.

6. Maintain the System

- Keep the knowledge base up-to-date by regular review and update.
- Cater for new interfaces with other information systems, as those systems evolve.

Benefits of Expert Systems

- **Availability :** They are easily available due to mass production of software.
- **Less Production Cost :** Production cost is reasonable. This makes them affordable.
- **Speed :** They offer great speed. They reduce the amount of work an individual puts in.
- **Less Error Rate :** Error rate is low as compared to human errors.
- **Reducing Risk :** They can work in the environment dangerous to humans.
- **Steady response :** They work steadily without getting motional, tensed or fatigued.

Machine Learning- Data Distribution, Machine Learning Process, Tools, TensorFlow

Machine Learning (ML) :

Machine Learning (ML) is a type of **Artificial Intelligence (AI)** that allows computers to learn from data without being explicitly programmed. In other words, **ML enables machines to get better at tasks over time by analyzing patterns in data.**

Key Points About Machine Learning (ML)

1. **Learning from Data**:
   - **ML** helps computers learn from past data so they can make predictions or decisions in the future.
   - **Example**: A system that predicts the weather by analyzing past weather data (temperatures, humidity, etc.).
2. **Improvement Over Time**:
   - As the system is exposed to more data, it can improve and become more accurate without being reprogrammed.
   - **Example**: A recommendation system on Netflix gets better at suggesting movies as it learns more about what you like.
3. **Pattern Recognition**:
   - ML helps computers identify patterns in large sets of data, which they can use to make decisions.
   - **Example**: An email spam filter learns to recognize patterns in email content and can automatically sort spam emails.
4. **Types of Machine Learning**:
   - **Supervised Learning**: The system is trained on labeled data (data with correct answers) to make predictions or decisions.
     - **Example**: Training a model to recognize pictures of cats and dogs by showing it many labeled images (cat or dog).
   - **Unsupervised Learning**: The system is given data without labels and tries to find patterns or groupings by itself.
     - **Example**: Grouping customers based on buying behavior without knowing beforehand what categories (or labels) to use.

    - **Reinforcement Learning**: The system learns by interacting with its environment and receiving rewards or penalties based on its actions.
        - **Example**: A robot learning to play a video game by receiving points (reward) for actions that help it score.
5. **Prediction and Decision Making**:
    - Machine Learning is used for making predictions or decisions based on new, unseen data.
    - **Example**: A bank using ML to predict if someone will pay back a loan based on past financial history.
6. **Automation**:
    - ML automates tasks that would normally require human intelligence, saving time and effort.
    - **Example**: Self-driving cars use ML to analyze their surroundings and make driving decisions.
7. **Real-time Learning**:
    - Machine Learning can be used for real-time decisions, improving instantly as new data arrives.
    - **Example**: An online streaming service adapting its recommendations in real-time based on what you are currently watching.
8. **Accuracy**:
    - The more data the system receives, the better its predictions become. However, accuracy depends on the quality of the data.
    - **Example**: A medical AI system that becomes more accurate at diagnosing diseases as it processes more patient data.
9. **Applications**:
    - **ML** is used in various fields like healthcare, finance, marketing, and entertainment.
    - **Example**: In healthcare, ML is used to analyze medical images and help doctors diagnose diseases such as cancer.
10. **No Explicit Programming**:

- ML models learn to make decisions without being specifically programmed to do so, as they adapt based on data.
- **Example**: An e-commerce site uses ML to recommend products based on your browsing history without being specifically programmed to suggest those items.

Example of Machine Learning in Action:

1. **Spam Filters**: A spam filter that learns to identify junk emails by analyzing thousands of emails and recognizing patterns that suggest whether an email is spam or not.
2. **Voice Assistants**: Voice assistants like Siri and Alexa use ML to understand your voice commands and improve their responses over time by learning from your interactions.
3. **Fraud Detection**: Banks use ML to detect fraudulent transactions by analyzing patterns in your spending habits and flagging unusual activity.

Machine Learning- Data Distribution :

Data Distribution in Machine Learning refers to how the data points (or values) are spread out across the dataset. It shows the pattern or structure of the data, helping machine learning models understand how data behaves and make accurate predictions.

Key Points About Data Distribution in Machine Learning

1. **What is Data Distribution?**
   - Data distribution describes how data values are arranged or distributed in a dataset. It tells us how often certain values occur and whether they are spread out or clustered in certain areas.
   - **Example**: In a dataset of people's ages, if most people are between 20 and 30 years old, the data distribution is concentrated around that age range.
2. **Why is Data Distribution Important?**
   - Understanding the data distribution helps machine learning models make better decisions. If the data is uneven or skewed, the model might make errors.
   - **Example**: If a dataset for predicting house prices has mostly data on small houses, the model might not predict large house prices well.
3. **Types of Data Distribution**:
   - **Uniform Distribution**: Every data point has an equal chance of occurring.
     - **Example**: Rolling a fair die, where each number (1-6) has the same probability of showing up.
   - **Normal Distribution**: Data is spread in a bell curve, where most values are close to the average, and fewer values are far from it.

        - **Example**: Heights of people in a population often follow a normal distribution (most people are of average height).
    - **Skewed Distribution**: Data is not evenly distributed. One side of the data is more spread out than the other.
        - **Example**: Income distribution where most people earn average wages, but a few people earn extremely high incomes.
4. **Real-Life Example**:
    - **Example**: If you are trying to predict someone's exam score based on the amount of time spent studying, the data distribution could be normal if most people studied a similar amount of time, and only a few studied much more or less.
5. **Data Imbalance**:
    - If a dataset has more examples of one class than another, it is called **imbalanced**. This can lead to inaccurate predictions because the model will learn more about the majority class.
    - **Example**: In a dataset of emails, there might be far more "non-spam" emails than "spam" emails. This imbalance could make the spam filter less accurate at detecting spam.
6. **Training and Testing Data**:
    - Data distribution affects how the training and testing data are split. If the training data doesn't represent the real-world distribution, the model may not perform well.
    - **Example**: If a model is trained on data from one region and tested on data from another region, the distribution might differ, causing the model to make inaccurate predictions.
7. **Feature Distribution**:
    - Each feature (or column) in a dataset can have its own distribution. Understanding the distribution of each feature helps determine which algorithms to use.
    - **Example**: If a feature like "age" in a dataset has a normal distribution, algorithms like linear regression might work well. However, if the data is heavily skewed, other methods might be more suitable.
8. **Visualizing Data Distribution**:
    - Data distribution can be visualized using graphs like histograms, box plots, or density plots. These help us understand how the data is spread.
    - **Example**: A histogram showing the distribution of ages in a dataset will show how many people fall into each age group.
9. **How Data Distribution Affects Models**:

- Some algorithms assume certain data distributions. For example, linear regression assumes a normal distribution of errors. If the data doesn't match the assumptions, the model might not perform well.
- **Example**: If you try to use linear regression on skewed data without transforming it, the model may not make accurate predictions.

10. **Data Transformation**:

- Sometimes, we can transform the data to change its distribution to make it easier for machine learning models to work with.
- **Example**: If a feature like income has a skewed distribution, you might apply a logarithmic transformation to make it more normal, which improves model performance.

Machine Learning Process :

The **Machine Learning (ML) process** involves several steps to build a model that can learn from data, make predictions, and improve over time. Here's a simple breakdown of the process:

1. Define the Problem

- **What it is**: Identify the specific problem you want to solve with ML. This could be anything from predicting prices to classifying images.
- **Example**: You want to predict the price of houses based on features like size, location, and number of rooms.

2. Collect Data

- **What it is**: Gather the data that will help your machine learning model learn. This data is crucial as it is the foundation of the model.
- **Example**: You collect a dataset of past house prices, with details such as square footage, number of bedrooms, neighborhood, etc.

3. Clean and Prepare Data

- **What it is**: Before feeding data into the model, it needs to be cleaned and preprocessed. This means fixing missing values, removing outliers, and converting data into a useful format.
- **Example**: If some houses have missing values for the number of bedrooms, you might replace them with the average number of bedrooms from other houses in the dataset.

4. Choose a Model

- **What it is**: Select an appropriate machine learning algorithm to solve your problem. Different algorithms are used for different types of tasks (e.g., classification, regression).
- **Example**: If you want to predict house prices, you might use a **linear regression** model, which predicts continuous values.

5. Train the Model

- **What it is**: The model is trained using the data you've collected. During training, the model learns patterns in the data to make predictions.
- **Example**: The linear regression model learns how house features (size, location) relate to prices by finding patterns in the training data.

6. Test the Model

- **What it is**: After training, the model is tested on new, unseen data to see how well it performs.
- **Example**: You test the trained model on a new set of houses and check how close the predicted prices are to the actual prices.

7. Evaluate the Model

- **What it is**: You evaluate how well the model performs using metrics like accuracy, precision, or mean squared error. This helps you understand how good the model is at making predictions.
- **Example**: In house price prediction, you might calculate the **mean squared error (MSE)**, which measures how far off your predicted prices are from the actual prices.

8. Improve the Model

- **What it is**: Based on the evaluation, you might adjust the model to improve its accuracy. This could include adding more data, changing the algorithm, or fine-tuning parameters.

- **Example**: If the model isn't predicting house prices well, you might try using a more complex model like **decision trees** or **neural networks**.

9. Deploy the Model

- **What it is**: Once the model is accurate enough, it's deployed into a real-world environment where it can make predictions on new data.
- **Example**: You deploy your house price prediction model to a website where users can input house details and get an estimated price.

10. Monitor and Maintain the Model

- **What it is**: The model's performance should be monitored over time to ensure it continues to work well. If it starts to perform poorly, you may need to retrain it with new data.
- **Example**: If new data about house prices is available, you can retrain the model to keep it up to date with the latest trends.

Summary of the Machine Learning Process:

Step	Description	Example
1. Define the Problem	Identify the problem to solve.	Predict house prices based on features like size, location.
2. Collect Data	Gather relevant data for training the model.	Collect data on past house sales, including features and prices.
3. Clean and Prepare Data	Clean the data by handling missing values and formatting it properly.	Fix missing data, remove outliers, and standardize units.
4. Choose a Model	Select the right machine learning algorithm for the task.	Use linear regression to predict house prices.

Step	Description	Example
5. Train the Model	Train the model using the collected and prepared data.	The model learns relationships between house features and prices.
6. Test the Model	Test the trained model on unseen data to check performance.	Test on new house data to see if predictions match actual prices.
7. Evaluate the Model	Evaluate the model's accuracy using metrics.	Check the mean squared error (MSE) of the predictions.
8. Improve the Model	Adjust the model based on evaluation results.	Try using decision trees for better accuracy.
9. Deploy the Model	Deploy the model for use in real-world applications.	Make the house price model available on a website for users.
10. Monitor and Maintain	Continuously monitor the model's performance and retrain if needed.	Retrain the model as new house price data becomes available.

Machine Learning Tools :

Machine learning tools are software, frameworks, or libraries that help you build, train, and deploy machine learning models. These tools simplify the process of applying machine learning techniques and make it easier for people to use ML in their work.

Key Points About Machine Learning Tools

1. **What are Machine Learning Tools?**
   - These are tools that help in creating, training, and evaluating machine learning models.
   - **Example**: Scikit-learn, TensorFlow, and PyTorch are all examples of machine learning tools.
2. **Types of Machine Learning Tools**:
   - **Libraries**: Pre-built collections of functions and classes to perform machine learning tasks.
   - **Frameworks**: Tools that provide a more structured way to design and deploy models.
   - **Integrated Development Environments (IDEs)**: Tools that provide an environment to write, test, and debug machine learning code.

Common Machine Learning Tools

Tool	Description	Example Use Case
Scikit-learn	A simple and easy-to-use library for basic machine learning tasks like classification, regression, clustering, etc.	Used to build simple models like linear regression or decision trees.
TensorFlow	A powerful library developed by Google for deep learning and neural networks.	Used for building complex models like neural networks and image recognition systems.

Tool	Description	Example Use Case
Keras	A high-level framework built on top of TensorFlow that makes it easier to create neural networks.	Used to quickly prototype deep learning models.
PyTorch	A deep learning framework developed by Facebook, similar to TensorFlow, but more flexible.	Used in academic research and for building deep learning models in computer vision and NLP.
XGBoost	An optimized library for boosting algorithms, great for structured/tabular data.	Used to build models for structured data (e.g., predicting customer churn).
Pandas	A data manipulation library used to clean and prepare data for machine learning.	Used to clean, manipulate, and prepare data for modeling (e.g., handling missing values, merging datasets).
Matplotlib/Seaborn	Visualization libraries that help in plotting data and results.	Used to create graphs and plots to understand data or evaluate model performance (e.g., scatter plots, histograms).
Jupyter Notebooks	A popular tool that allows you to write and run machine learning code in an interactive environment.	Used for writing code, analyzing data, and documenting results all in one place.
Google Colab	A cloud-based version of Jupyter Notebooks, often used for free access to GPUs and faster training.	Used for training models on cloud servers with GPU support, especially in deep learning tasks.

Tool	Description	Example Use Case
Azure ML	A cloud-based service by Microsoft that helps in building, training, and deploying machine learning models.	Used to train models on the cloud and deploy them as web services.

Examples of How Machine Learning Tools Are Used

1. **Scikit-learn**:
   - **Task**: Building a model to predict whether a customer will buy a product based on their browsing history.
   - **How**: Use **Scikit-learn** to train a logistic regression model or decision tree on customer data.
2. **TensorFlow**:
   - **Task**: Building a deep learning model to recognize objects in images.
   - **How**: Use **TensorFlow** to design and train a Convolutional Neural Network (CNN) for image classification.
3. **Keras**:
   - **Task**: Quickly prototyping a neural network to predict house prices based on features like size, location, and year built.
   - **How**: Use **Keras** to create a deep learning model and fine-tune it for accuracy.
4. **PyTorch**:
   - **Task**: Implementing a model that generates text based on previous examples (e.g., chatbot).
   - **How**: Use **PyTorch** to train a Recurrent Neural Network (RNN) to generate text sequences.
5. **XGBoost**:
   - **Task**: Predicting customer churn for a subscription service based on features like age, usage, and subscription type.
   - **How**: Use **XGBoost** to build a strong, gradient-boosting model to predict whether a customer will churn.
6. **Pandas**:
   - **Task**: Preparing and cleaning data for use in a machine learning model.
   - **How**: Use **Pandas** to remove missing values, handle categorical data, and normalize numerical features.
7. **Matplotlib/Seaborn**:

- **Task**: Visualizing the results of your machine learning model to better understand performance.
- **How**: Use **Matplotlib** to plot confusion matrices or **Seaborn** to visualize feature importance.

8. **Jupyter Notebooks**:
   - **Task**: Writing and testing machine learning code interactively.
   - **How**: Use **Jupyter Notebooks** to write Python code for data analysis, training models, and plotting results in one document.
9. **Google Colab**:
   - **Task**: Training a deep learning model on a large dataset without needing your own high-powered computer.
   - **How**: Use **Google Colab** to access free GPUs and train a neural network on image data.
10. **Azure ML**:
    - **Task**: Deploying a machine learning model to predict product demand in real-time.
    - **How**: Use **Azure ML** to deploy your trained model as a web service, allowing users to query predictions over the internet.

Tensor Flow :

Machine Learning - TensorFlow in Simple Words

TensorFlow is a popular open-source library developed by Google that helps build and train machine learning models, particularly deep learning models. It is widely used for tasks involving large datasets and complex algorithms, like image recognition, natural language processing, and more.

Key Points About TensorFlow

1. **What is TensorFlow?**
   - TensorFlow is a library used for building machine learning models, particularly deep learning models. It helps with creating neural networks to solve complex problems.

    - **Example**: TensorFlow is used to build models that can recognize cats in pictures or translate text from one language to another.
2. **How Does TensorFlow Work?**
    - It uses **tensors** (multi-dimensional arrays) for data representation and computations. It performs calculations by using a computational graph where each operation is a node, and data flows between them.
    - **Example**: If you're training a model to recognize images, TensorFlow breaks down the process into steps, like converting images into numbers, applying mathematical operations, and making predictions.
3. **Deep Learning with TensorFlow:**
    - TensorFlow is widely used for **deep learning**, which involves using neural networks with many layers (deep neural networks) to solve problems like image classification and speech recognition.
    - **Example**: A deep learning model in TensorFlow might recognize handwritten digits (like the ones in postal codes) or detect objects in images.
4. **Automatic Differentiation:**
    - TensorFlow automatically computes gradients during the training of models. This is important for updating the weights of the neural network using optimization techniques like **gradient descent**.
    - **Example**: In a model predicting house prices, TensorFlow adjusts the weights to minimize the error in predictions over time.
5. **TensorFlow for Different Tasks:**
    - It can be used for a variety of tasks:
        - **Classification** (e.g., classifying images of animals).
        - **Regression** (e.g., predicting house prices).
        - **Natural Language Processing (NLP)** (e.g., translating text).
        - **Reinforcement Learning** (e.g., teaching an agent to play a game).
    - **Example**: Using TensorFlow to predict stock prices or teach an AI to play chess.
6. **Keras and TensorFlow:**
    - **Keras** is a high-level API that makes building models in TensorFlow easier. It's user-friendly and simplifies many processes like defining neural networks and training them.
    - **Example**: In TensorFlow, you can use Keras to easily define and train a neural network model for image classification.
7. **Deployment:**
    - TensorFlow also provides tools to **deploy** machine learning models into real-world applications, allowing them to run on devices (like smartphones) or on the cloud.

- **Example**: After training a model to detect faces in images, you can use TensorFlow to deploy this model on an app that can detect faces from photos in real-time.

8. **TensorFlow Lite:**
   - TensorFlow Lite is a lightweight version of TensorFlow that is optimized for mobile and embedded devices.
   - **Example**: Running a machine learning model directly on a smartphone to recognize objects in photos, without needing internet access.
9. **TensorFlow.js:**
   - TensorFlow.js allows you to run machine learning models in the browser or on Node.js, using JavaScript.
   - **Example**: You can create a web application that predicts the weather by training and running a model in the browser.
10. **TensorFlow Hub:**
    - A repository of pre-trained machine learning models that you can reuse and fine-tune for your own tasks.
    - **Example**: You can use a pre-trained model from TensorFlow Hub to detect objects in images, saving you time in training from scratch.

Example of Using TensorFlow

Let's say you want to train a machine learning model to recognize handwritten numbers (like in the MNIST dataset).

1. **Load the Data**:
   - You first load the MNIST dataset (images of handwritten digits) into TensorFlow.
2. **Create the Model**:
   - Using **Keras** (part of TensorFlow), you define a neural network with layers (e.g., input layer, hidden layers, and output layer).
3. **Train the Model**:
   - The model learns by adjusting weights using the data, applying **backpropagation** and **gradient descent**.
4. **Test the Model**:
   - After training, you test the model with new handwritten numbers to see how accurately it can predict the digits.
5. **Use the Model**:
   - Finally, you can use the trained model to classify new images of handwritten digits in real-world applications.

Summary of TensorFlow Features:

Feature	Description	Example Use Case
Tensors	Multi-dimensional arrays used for data and calculations.	Representing data for image classification or text processing.
Deep Learning	Used for building and training neural networks.	Image recognition or speech-to-text applications.
Keras API	High-level API that simplifies the model-building process.	Quick prototyping of deep learning models.
Gradient Descent	Automatically adjusts weights using gradients to improve the model.	Training models for accurate predictions (e.g., price prediction).
TensorFlow Lite	Optimized for mobile and embedded devices.	Running AI models on smartphones or embedded systems.
TensorFlow.js	Running machine learning models directly in the browser with JavaScript.	Web-based AI applications like live object detection.
TensorFlow Hub	A repository of pre-trained models to use or fine-tune.	Using a pre-trained model for image classification.